



POLITECNICO
MILANO 1863

ASSIGNMENT 1

STRUCTURAL DYNAMICS AND AEROELASTICITY

Numerical Code

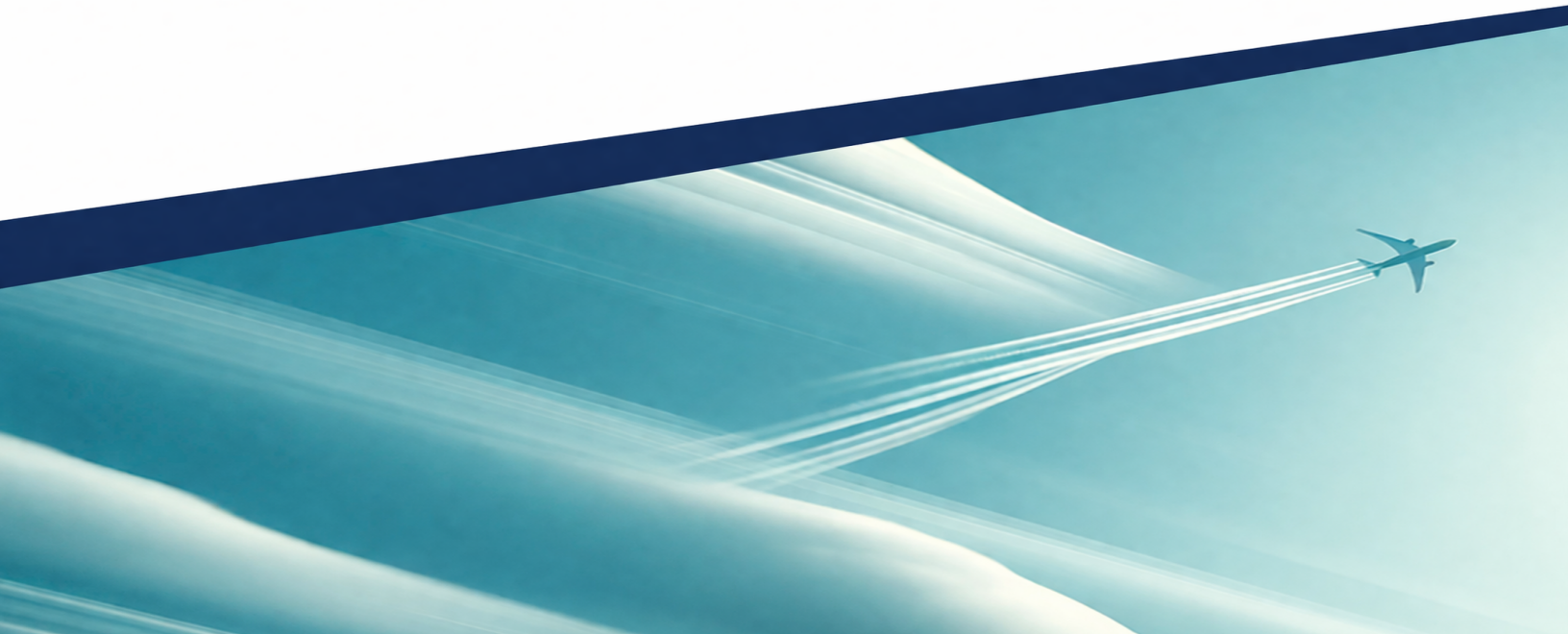
POLITECNICO DI MILANO
Laurea Magistrale – Aeronautical Engineering

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Overview

This document provides the execution instructions for the numerical code submitted for Assignment 1. The numerical work is split into two MATLAB files :

- `Camile_BRIARD_aeroelastic_assign1_notebook.mlx` : results for Questions 1 to 5;
- `Camile_BRIARD_q6_assign1.m` : result for Question 6.

Software

- Software used : MATLAB
- MATLAB release : **R2025b**
- Full version string : **25.2.0.2998904**

Required Libraries and Dependencies

No external library, package, or user-defined dependency is required outside MATLAB itself. The submitted files use only standard MATLAB functionality, including Live Script support for the `.mlx` notebook.

Execution Instructions

Questions 1 to 5

File to run : `Camile_BRIARD_aeroelastic_assign1_notebook.mlx` Recommended execution procedure :

1. Open MATLAB.
2. Set the repository root as the current folder.
3. Open the file `Camile_BRIARD_aeroelastic_assign1_notebook.mlx`.
4. Run the full Live Script with `Run All`.

Expected runtime :

- approximately 1 minute on a machine equivalent to an Intel i9 with 16 GB RAM;
- allow about 1 to 3 minutes for full execution.

During execution, the notebook computes the numerical results and generates the figures used in the report. The notebook file is intentionally submitted as a Live Script because it contains both commented explanations and numerical sections for Questions 1 to 5. Do not take into account the error about `ME.message`, it does not affect the execution and the results.

Question 6

File to run : `Camile_BRIARD_q6_assign1.m` The notebook and the script are independent and can be executed separately. Recommended execution procedure :

1. Open MATLAB.
2. Set the repository root as the current folder.
3. Open the file `Camile_BRIARD_q6_assign1.m`.
4. Open the `Command Window` to view printed outputs.
5. Run the script.